

The three Shilluk right-edge H's: An argument for subtonal features and strata*

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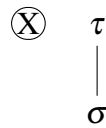
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1. Introduction

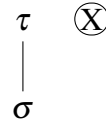
In a recent analysis of grammatical tone, i.e. tonal alternations that only manifest in specific morphosyntactic contexts, Trommer (2024) demonstrates that morpheme-specific phonology is not required to derive the phenomenon. He proposes to reanalyze it as the interaction between general phonological constraints on tonal realization and the way tonal exponents are concatenated with their morphological base. For example, tonal exponents consisting of the same tonal string can induce different tonological effects when one is linearized as a prefix (1a) and the other as a suffix (1b). Additionally, tonal strings linearized identically with respect to their base can still result in different phonological outcomes if one is monomorphemic (2a) and the other, polymorphemic (2b).

(1) Tonal affix linearization

a. Tonal prefix

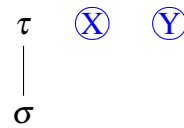


b. Tonal suffix



(2) Tonal string structure

a. Single exponent



b. Multiple exponents



Under this approach, two different exponents consisting of the same tonal string with the same internal complexity and realized on the same edge of their base should trigger identical grammatical tone effects, a prediction which appears to be challenged empirically. In the inflectional paradigm of nouns in Shilluk (Western Nilotic, South Sudan), three morphological categories – the plural pensive (PERT.PL), the associative plural (APL) and

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the vocative (VOC) – are realized as a single H suffix, and yet produce different outcomes in the same tonal environment, as illustrated below in (3).

- (3) The Shilluk H suffixes (Remijsen and Ayoker 2019, 2021)
- | | | | PERT.PL | | APL | | VOC | | |
|----|----------|-----|---------|-----|--------|-----|--------|------|--------------|
| a. | kɔ̃t/kul | L | kɔ̃t | LH | kul | LH | kul | LH | ‘rain’/‘Kul’ |
| b. | bo::tɔ | L.L | bo::tɪ | L.H | bo::tɪ | L.M | bo::tɔ | L.LH | ‘craftsman’ |

While these exponents systematically surface as H in the context of monosyllabic L nouns, they all behave differently when are suffixed to bisyllabic L.L nouns: the APL H lowers to M, the PERT.PL H replaces the final syllable L without lowering, and the VOC H fails to replace the final tone or lower.

In this paper, I argue that the Shilluk pattern does not represent a real counterexample to Trommer’s (2024) approach once two further theoretical assumptions are made: first, a stratal architecture for the phonological component (Kiparsky 2000, Bermúdez-Otero 2018), and second, the decomposition of tone into a feature geometry (Snider 1999). More concretely, I propose that the APL, PERT.PL and VOC H’s are all introduced at different strata, deriving their different behaviour, and that the stratum at which APL is introduced enforces register l-sharing across syllable boundaries, deriving the limited scope of H-lowering.

The remainder of the paper is structured as follows. In section 2, I present additional information on the language and the APL, PERT.PL and VOC patterns. In section 3, I detail my theoretical assumptions and present a Stratal OT analysis of the patterns. Section 4 briefly discusses the implications of the proposal and concludes.

2. Background

In this section, I first provide additional information on the tonology and nominal morphology of Shilluk (section 2.1), before turning to the APL (2.2), PERT.PL (2.3) and VOC patterns (2.4), respectively. All the data on which I report are the result of Bert Remijsen and Otto Gwader Ayoker’s extensive fieldwork on the language, for which I am incredibly grateful.

2.1 Tone and nominal morphology

The Shilluk tone system displays three tonal heights: L(ow), M(id), and H(igh) (Remijsen et al. 2011). Contour tones are restricted to heavy syllables, i.e. CV(::)C, (C)V:, and belong to the falling (HM, HL, ML) or rising (LH, MH) categories (Remijsen and Ayoker 2018). Additionally, morphologically derived environments allow for the creation of two additional types of contour tones: two late-aligned falls (HXL, MXL) and a dipping tone (HLH) (Remijsen and Ayoker 2019, 2021).

On top of their base form (BF), nouns inflect for possession (PERT.SG, PERT.PL), general modification (CS), proximal deixis (PROX), associative plurality (APL) and vocative case (VOC). These inflectional categories are essentially realized via grammatical tone al-

ternations, whose outcomes are illustrated in the table below in (4) for the monosyllabic noun *gwok* ‘dog’ and the bisyllabic noun *bɔ::tɔ* ‘craftsman’.

(4) Tonal inflection in Shilluk nouns				(Remijsen and Ayoker 2019, 2021)	
BF	gwok	ML		bɔ::tɔ	L.L
PERT.SG	gwo::k	HM		bɔ::tɪ	L.L
PERT.PL	gwo::k	H		bɔ::tɪ	L.H
CS	gwo::ŋ	HM		bɔ::nɪ	L.L
PROX	gwo::ŋ	HXL		bɔ::nɪ	L.L
APL	gwo::k	H		bɔ::tɪ:	L.M
VOC	gwok	MH	‘dog’	bɔ::tɔ:	L.LH ‘craftsman’

Moreover, APL and VOC stand out in that their realization depends on the internal make-up of the nominal expression: they invariably surface on the rightmost word, whether it is a noun or not, as illustrated in (5a) and (5b) below for the APL and the VOC, respectively. This being said, their tonal behaviour appears to be constant across all contexts, and so I will abstract away from this complication in what follows.

- (5) APL and VOC appear on the rightmost word in the nominal expression
- a. ja:^{HM} bɔ::nɪ^{L.L} a^L tɛ:k^{MH}
PRT craftsman REL strong.APL
‘associates of the strong craftsman’ (Remijsen and Ayoker 2021:58)
- b. bɔ::nɪ^{L.L} a^L tɛ:k^{MH}
craftsman REL strong.VOC
‘strong craftsman!’ (Remijsen and Ayoker 2021:70)

2.2 The associative plural (APL)

The realization of the APL H is subject to a $H \sim M$ alternation. The H alternant appears in the context of all monosyllabic nouns, as well as with bisyllabic nouns whose final tone is H (6-a), whereas the M alternant surfaces in the context of bisyllabic nouns whose tonal melody ends in L or M (6-b). This suggests that the H alternant is basic, and that a process of H-lowering generates the M alternant.

(6) The APL pattern				(Remijsen and Ayoker 2021:60, 63)	
	BF		APL		
a.	kul	L	kul	LH	‘warthog’
	kwaɲ	M	kwa::ɲ	MH	‘porridge’
	tɔŋ	H	tɔŋ	H	‘spear’
	wat	ML	wa:t	MH	‘relative’
	gwok	ML	gwo::k	H	‘dog’
	ɲa::lɔ	ML.H	ɲa::lɪ:	ML.H	‘python’
b.	bɔ::tɔ	L.L	bɔ::tɪ:	L.M	‘craftsman’
	ta:nɔ	M.M	ta:nɪ:	M.M	‘roof’
	cɔlɔ	HL.L	cɔlɪ:	HM.M	‘Shilluk’
	ja::kɔ	HM.M	ja::kɪ:	HM.M	‘chief’

The APL inflection also uncovers a tonal contrast between nouns whose BF is ML: the nouns *wat* ‘relative’ and *gwok* ‘dog’ respectively surface as MH and H in the APL. I take this as evidence that the BF tonal shape must be derived in the course of the phonological derivation: if it were identical to the underlying tonal shape of nouns, it would keep track of the contrast between these two classes of ML nouns, contrary to fact.

2.3 The plural pertensive (PERT.PL)

Unlike what happens with the APL H, the realization of the PERT.PL H is consistently H across all tonal contexts, as illustrated below in (7).

(7) The PERT.PL pattern (Remijsen and Ayoker 2019:37, 38, 39, 41)

BF		PERT.PL		
a.	kɔ̌t	L	kɔ̌t	LH ‘rain’
	kʌc	M	kʌ::c	MH ‘hunger’
	du:p	H	du::p	H ‘rodent’
	tɔk	ML	tɔk	MH ‘edge’
	kwot	ML	kwo:t	H ‘shield’
	wutɔ	ML.H	wutɪ	ML.H ‘ostrich’
b.	bɔ:tɔ	L.L	bɔ:tɪ	L.H ‘craftsman’
	dje:rɔ	M.M	dje:rɪ	M.H ‘truth’
	tulɔ	HL.L	tulɪ	H.H ‘owl’
	ja:kɔ	HM.M	ja:rɪ	H.H ‘chief’

However, the PERT.PL inflection uncovers the same tonal contrast as the APL with respect to nouns whose BF is ML: the nouns *tɔk* ‘edge’ and *kwot* ‘shield’ respectively surface as MH and H in the PERT.PL. Also identical to the APL pattern is the fact that in the PERT.PL, the inflectional tone replaces the final tone a bisyllabic noun displays in its BF.

2.4 The vocative (VOC)

The realization of the VOC H is similar to that of the PERT.PL H in that it is consistently H across all tonal contexts, as illustrated below in (8).

(8) The VOC pattern (Remijsen and Ayoker 2021:70, 71, 73)

BF		VOC		
a.	kul	L	kul	LH ‘Kul’
	dɛ:ŋ	M	dɛ:ŋ	MH ‘Deng’
	twɔ:ŋ	H	twɔ:ŋ	H ‘Twong’
	wat	ML	wat	MH ‘relative’
	gwok	ML	gwok	MH ‘dog’
	ɲʌ:lɔ	ML.H	ɲʌ:lɔ:	ML.H ‘python’
b.	bɔ:tɔ	L.L	bɔ:tɔ:	L.LH ‘craftsman’
	cɔlɔ	HL.L	cɔlɔ:	HL.LH ‘Shilluk’
	ja:kɔ	HM.M	ja:kɔ:	HM.MH ‘chief’

Unlike the APL and PERT.PL patterns, however, the VOC inflection strictly follows the BF tonal contrasts: the nouns *wat* ‘relative’ and *gwok* ‘dog’, which both have a ML-toned BF, also both surface as MH in the VOC. Moreover, the VOC H never replaces the final BF tone in a bisyllabic noun: the nouns *cɔɔ* ‘Shilluk’ and *ja::kɔ* ‘chief’, whose BF shapes are HL.L and HM.M, respectively, surface as HL.LH and HM.MH in the VOC, with their final vowel lengthened.

2.5 Interim summary

The inflectional paradigm of Shilluk nouns contains three morphological categories whose exponent can be analysed as a H suffix: the APL, the PERT.PL and the VOC. The phonological properties of these various H suffixes can be captured using two criteria: ① whether the suffix participates in H-lowering; and ② whether the realization of the suffix pre-empts the BF tonal shape. The APL H both undergoes H-lowering and pre-empts the BF tonal shape. The PERT.PL H does not participate in H-lowering, but it pre-empts the BF tonal shape. The VOC H neither participates in H-lowering nor pre-empts the BF tonal shape.

These phonological properties reveal crucial interactions between morphological exponence and phonological processes. First, the PERT.PL and VOC H's not only fail to participate in H-lowering, they counterfeed it, meaning that they are introduced at a point in the derivation at which H-lowering has already applied and cannot reapply. Second, the APL and PERT.PL H's bleed the creation of the BF tonal shape, meaning that they are introduced at a point in the derivation at which their integration into the tonal structure eliminates the possibility to apply the process responsible for creating the BF tonal shape.

The H-lowering process also hints at additional properties of the Shilluk tone system. As things stand, the context for the process is stated in disjunctive terms, i.e. when a L *or* a M precedes the APL H across a syllable boundary. I reinterpret this disjunction as the fact that L and M exhibit natural class behaviour with respect to tone lowering, and that they accordingly share a property which enables this phonological grouping.

3. Analysis

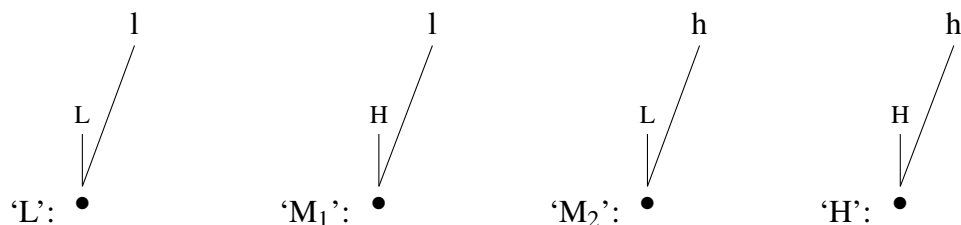
In this section I introduce two representational assumptions for my proposal, namely subtonal features and morphological colouring (3.1). I then present an analysis of the Shilluk APL, PERT.PL and VOC H's couched in Stratal OT. Key to this analysis is the idea that each exponent is introduced at a different stratum: the APL H is a stem-level exponent (section 3.2); the PERT.PL H is a word-level exponent (section 3.3); and the VOC H is a phrase-level exponent (section 3.4).

3.1 Theoretical assumptions

The fact that tones may exhibit natural class behaviour with respect to phonological processes has been seen as evidence that they do not stand as atomic units and should be broken down into subtonal features. In this paper I adopt Snider's (1999) approach that decomposes tone into two features, which is in line with other proposals in the literature (Yip

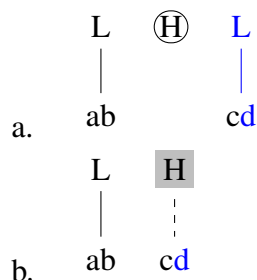
1980, McPherson 2017, Meyase 2021). In his approach, a ‘tone’ consists of a melody feature H or L, which identifies the height of a tone with respect to a point of reference (Higher or Lower), and a register feature h or l, which identifies the actual position of that point of reference (high or low). These features exist on independent autosegmental tiers, and their association to a T(one-)B(earing)U(nit) is mediated by a tonal root node (●). Combining these ingredients together allows for four featurally distinct, fully-specified tones, as illustrated below in (9).

(9) Tonal representations in Register Tier Theory Snider (1999)



The fact that the phonology is sensitive to the morphological structure of a given string has been seen as evidence that it should be directly encoded into the string. In this paper I adopt van Oostendorp's (2006) proposal that all autosegments belonging to the same exponent bear the same index, or “colour”. For example, the toy phonological string in (10a) below is made up of two exponents as its various elements are assigned to one of two colours. The toy example in (10b) also consists of two exponents, as indicated by the two colours in the representation. The H tone, by virtue of being epenthetic, does not belong to any exponent, and is accordingly colourless.

(10) Colours and colourlessness van Oostendorp (2006)



By allowing phonological constraints to refer directly to colour, it becomes possible to distinguish between underlying (i.e., coloured) and epenthetic (i.e., colourless) nature of a given autosegment in carrying out the phonological evaluation, as well as to the tautomorphemicity (i.e., single colour) or heteromorphemicity (i.e., multiple colours) of autosegments. Note that reference to colour does not entail reference to a *specific* colour: the phonology has access to the morphological structure of the string it evaluates, but not to the identity of the morphemes that make up said string.

3.2 The APL: stem-level $\textcircled{\text{H}}$

I analyse the APL H as a stem-level exponent. This consequently means that H-lowering is a property of the Stem Level grammar, correctly deriving the fact that both the PERT.PL and VOC H's do not undergo it: their introduction at the Word and Phrase Level strata, respectively, is too tardy for them to do so.

In what follows, I illustrate my proposal with nouns whose BF shape is L.L. This allows me to touch on all crucial aspects of the proposal while simplifying matters that pertain to the underlying tonal specifications of Shilluk nouns, which receive a more thorough treatment in Jolin (submitted). I assume that all bisyllabic nouns in Shilluk have a tonally unspecified final syllable, as explicated by the $L\sigma$ notation for the nouns under consideration here.

I propose that the association between TBUs and tones is regulated by constraints that require tonal root nodes to associate to TBUs (11a) and for TBUs to be associated to a tonal root node (11b). I also propose that the creation of association lines between tautomorphemic autosegments is generally disfavoured (11c). Finally, I model H-lowering using a constraint that requires adjacent syllables to share a l-register feature (11d).

- (11)
- TO- σ : Assign * for every ● not associated to a σ .
 - SPECIFY (SPEC): Assign * for every syllable that is not associated to a full tone (i.e., ●, T and t).
 - ALTERNATION (ALT): Assign * for every epenthetic association line between autosegments bearing the same colour.
 - SHARE-l: Assign * for every register-l feature that is not shared between two adjacent syllables.

I present the phonological evaluation of a L.L noun with the APL H in the tableau in (12). I abbreviate various faithfulness constraints as FTH for clarity purposes.

- (12) N-APL, Stem Level $L\sigma + \textcircled{\text{H}} \rightarrow \text{L.M}$

Input: a.	ALT	●-TO- σ	SHARE-l	FTH	SPEC
a.		*!	*		*
b.			*!	*	
c.				***	

The input candidate in (12-a) fails to integrate the APL exponent into the structure, which incurs a fatal $\bullet$ -TO- σ violation. Associating the APL exponent to the final, toneless syllable of the noun as in (12-b) circumvents a violation of both $\bullet$ -TO- σ and SPEC, but this still does not eliminate the SHARE-I violation. The winning candidate in (12-c) avoids violating it by additionally spreading the noun's l-register onto the APL, creating a {l,H} tone which is interpreted as [M]. The constraint ranking therefore ensures that all bisyllabic nouns whose tonal shape ends in a l-containing tone (i.e., L and M) will induce H-lowering on the APL H. It also correctly accounts for the fact that the APL exponent surfaces as H in all monosyllabic contexts, since they vacuously satisfy SHARE-I.

A final noteworthy property of the Stem Level grammar is the low ranking of SPEC, which allows for tonally underspecified outputs to be passed over to the Word Level. Indeed, there exists no phonological operation at the Stem Level that could satisfy SPEC while avoiding violations of higher-ranked constraints: spreading from the initial syllable is ruled out by ALT, and tone epenthesis is excluded by DEP- $\bullet$ (not shown here for reasons of space).

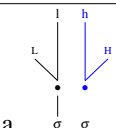
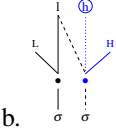
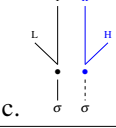
3.3 The PERT.PL: word-level $\textcircled{H}$

I analyse the PERT.PL H as a word-level exponent. The Word Level grammar differs from the Stem Level one in two important ways: first, FTH now outranks SHARE-I; and second, SPEC is now undominated and therefore outranks ALT.

I present the phonological evaluation of a L.L noun with the PERT.PL H in the tableau in (13).

(13) N-PERT.PL, Word Level

$$L\sigma + \textcircled{H} \rightarrow L.H$$

Input: a.	SPEC	$\bullet$ -TO- σ	ALT	FTH	SHARE-I
	*!	*			*
				***!	
				*	*

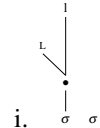
The input candidate in (13-a) fatally violates SPEC since it contains a toneless syllable. This is because the Stem Level grammar cannot enforce full tonal specification in the absence of a stem-level exponent. The reranking of FTH and SHARE-I derives the fact that the H-lowered candidate in (13-b) is no longer optimal at the Word Level, since it fares worse than the non-lowered candidate in (13-c) with respect to FTH.

A final noteworthy property of the Word Level grammar is the high ranking of SPEC, which strictly disallows tonally underspecified outputs to be passed over to the Phrase Level. Satisfying SPEC at the Word Level is what is responsible for creating the BF tonal shape. By introducing exponents at the Stem and Word Level respectively, APL and PERT.PL satisfy SPEC by docking onto the available syllabic space, which explains why they pattern together in pre-empting the realization of the BF tonal shape. In the absence of any additional exponent by the Word Level, satisfying SPEC must resort to carrying out phonological operations that violate lower-ranked constraints like ALT or DEP-•.

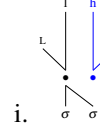
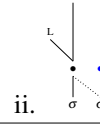
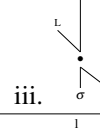
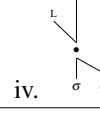
3.4 The VOC: phrase-level $\textcircled{\text{H}}$

I analyse the VOC H as a phrase-level exponent. I present the phonological evaluation of a L.L noun with the VOC H in the tableau in (14b).

(14) a. N, Word Level $L\sigma \rightarrow L.L$

Input: a.	SPEC	•-TO- σ	ALT	FTH	SHARE-I
i. 	*!				
ii. 			*	*	

b. N-VOC, Phrase Level $L.L + \textcircled{\text{H}} \rightarrow L.LH$

Input: a.	MAX-AL	•-TO- σ	ALT	*LIGHTC	DEP- μ
i. 		*!			
ii. 	*!				
iii. 				*!	
iv. 					*

The derivation of the VOC forms is more complex, as they must form pass through the Word Level grammar. Since the VOC H is a phrase-level exponent, it is necessarily absent from the Word Level evaluation, which is presented in (14a). As discussed above, all Word Level outputs display full tonal specification due to a high-ranking SPEC. This is why the

input candidate in (14a-i), which contains a toneless syllable, is ruled out. The winning candidate in (14a-ii) avoids a SPEC violation by spreading the available tone onto the toneless syllable, incurring a tolerable ALT violation. At the Phrase Level, the VOC exponent is concatenated with a fully specified noun, which creates a conflict for tonal realization. Associating the VOC H in a way that dislodges the final tone of the noun as in (14b-ii) incurs a fatal violation of MAX-AL, which penalizes the deletion of already existing association lines between autosegments. Preserving this association line as in (14b-iii) creates the correct tonal shape, but fails to derive the fact that all bisyllabic nouns display a long final vowel in the vocative. I propose that (14b-iii) can be excluded on the basis of the fact that it creates a contour tone on a light syllable, which is otherwise unattested in the language. The winning candidate (14b-iv) avoids violating *LIGHTC by epenthesisizing a mora, which derives the final lengthening effect.

4. Conclusion

In this paper, I presented an analysis of three morphological categories from the inflectional paradigm of Shilluk nouns which appeared to present an empirical challenge to Trommer's (2024) approach to grammatical tone effects. Indeed, the APL, PERT.PL and VOC, despite all representing instances of H-suffixation to a nominal base, produce different outcomes when concatenated to L.L nouns. I argued that this empirical challenge can be resolved if one assumes that these exponents are introduced at different phonological strata, and showed that the phonological behaviour of the APL can be elegantly captured under a featural approach to tonal representations. The analysis of Shilluk H-suffixation that I propose is therefore compatible with a more restrictive approach to grammatical tone effects. Additionally, it shows that both strata and enriched tonal representations may be necessary to derive grammatical tone effects in a single language, where previous work had only identified need for one or the other (Trommer 2021, 2022).

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